

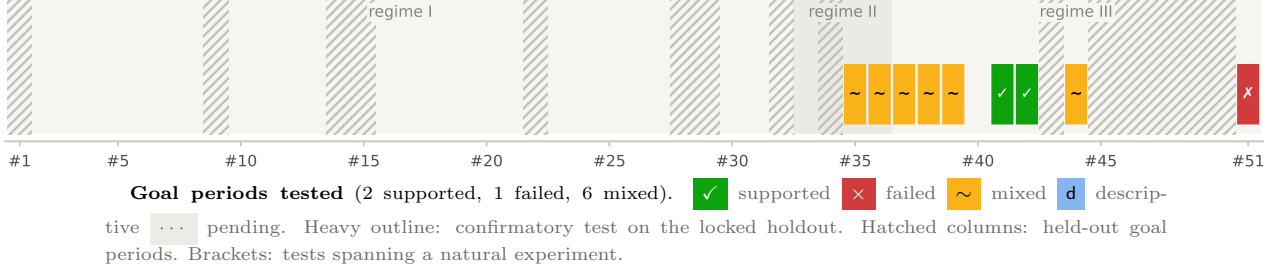
# H102: Domain walls between rooms, set by bridging agents

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 45% Faithfulness 1.5/5 Usefulness 2/5

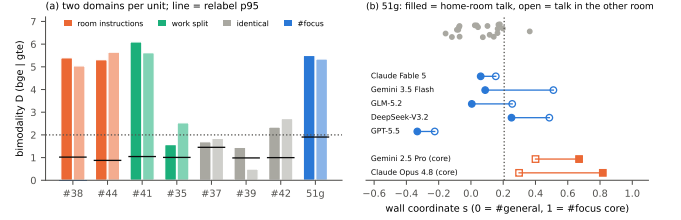
**Bottom line.** Rooms that do different work form two content domains separated by a sharp, empty wall; hoppers switch domains with the room they speak in and carry nothing home, so bridging agents do not widen the wall.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



## WHAT HAPPENED

HH332 says that rooms doing different things form two content domains. Agents who hop between rooms would sit inside the wall, and the wall would widen with their cross-room read-outs. H102 projects style-residualized content on the inter-room axis in 9 non-holdout units (#35–#44 and the #focus weeks of #51). It follows the five #focus hoppers and one #36 hopper. **The domains are real; the wall is not.** Where rooms do different work, stayers split into two groups ( $D$  5.3–6.1; relabel 95th percentile  $\approx 1$ ). Hoppers stay inside their home domain. The one frequent hopper sits at  $s = 0.06$  against a stayers' 95th percentile of 0.21. Hopping read-outs have no effect ( $\kappa_R - 0.014 [-0.042, 0.024]$ ). A hopper's talk follows the room it is speaking in. HH332's kill rule fires. The holdout is not run.



**Observables.** (a) Bimodality  $D$  per unit (bge solid, gte light); the black line is the relabel 95th percentile with the axis refitted. Instruction weeks, #41 and the #focus room separate most. (b) #51g wall coordinate. Grey: #general stayers. Blue: hoppers' home-room talk (filled) and their talk inside #focus (open). Orange: the two #focus residents' own-room and #general talk. No agent's home-room talk sits in the wall.

## SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

### PHYSICAL MODEL

Model 11 (kinetic soft-spin,  $n = 32$ ) read along one axis like an Ising order parameter (model 01). Agent content  $\mathbf{x}_i$  relaxes toward the content it reads, with weights  $\pi_{ij}$  from the context ledger, and toward its domain field  $\mathbf{h}_D$  (room work or instructions). The wall coordinate  $s$  is 0 at the home-domain centroid and 1 at the other domain.  $p_i$  is the share of read items from the other domain. Coupling predicts  $s \approx \kappa p$ ; a room-local field predicts that  $s$  jumps with the room of the statement.

$$\mathbf{x}_i \leftarrow (1 - \lambda)\mathbf{x}_i + \lambda[w \sum_j \pi_{ij} \mathbf{x}_j + (1 - w)(\mathbf{h}_{D(i)} + \mathbf{a}_i)] + \boldsymbol{\eta}$$

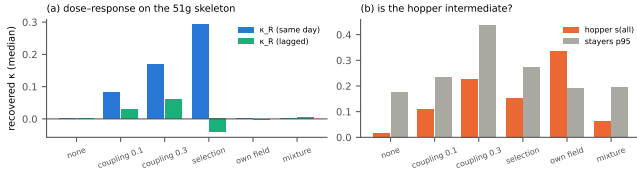
$$s_i = \frac{(\mathbf{x}_i - \mathbf{c}_{\text{home}}) \cdot \mathbf{u}}{(\mathbf{c}_{\text{other}} - \mathbf{c}_{\text{home}}) \cdot \mathbf{u}}, \quad s_i \approx \kappa p_i$$

$$D = \frac{\bar{z}_B - \bar{z}_A}{\sqrt{(\sigma_A^2 + \sigma_B^2)/2}}$$

H102-room-domain-walls · generated 2026-10-04 14:21 by in-  
fra/summaries/build\_summaries.py · card hypotheses/H102-room-domain-  
walls/README.md · ratings: H102 agent (suggested)

## RESULTS VS PREDICTIONS

| Prediction (locked)                       | Observed                                                        | Verdict |
|-------------------------------------------|-----------------------------------------------------------------|---------|
| P0 synthetic: $D$ size, $\kappa_R$ power  | $D$ size 0; $\kappa_R$ power $\geq 0.93$ , size 0.05–0.18       | mostly  |
| P1 $D \geq 2$ in #38, #44, #35, #41       | 5.40, 5.31, 1.57 (gte 2.53), 6.10; mostly identical median 1.70 |         |
| P2 hoppers intermediate                   | Table 5 0.06 vs stayers p95 0.21; failed none above             |         |
| P3 $\kappa_R > 0$ , $> \kappa_U$ , lagged | −0.014 [−0.042, 0.024]; lag failed −0.004                       | failed  |
| P4 wall width grows with reads            | Spearman 0.30 ( $p$ 0.43); weak #38/#44 $I$ 0.07/0.17           |         |
| P5 axis = kickoff direction               | cos  0.02 ( $p$ 0.95), 0.27 ( $p$ 0.14) failed                  | failed  |
| P6 #36 hopper shift $\geq 0.3$            | $\geq 0.25$ (gte 0.20); back to −0.13 after                     | mixed   |
| HH332 kill rule                           | hoppers not intermediate                                        | failed  |



**Second figure.** Synthetic worlds on the real #51g skeleton (40 each). (a) Coupling gives a same-day and a lagged  $\kappa_R$ . Selection gives a same-day  $\kappa_R$  only. An own-field world gives none. (b) Only the own-field world makes hoppers intermediate without a dose.

## GOAL PERIODS IN DETAIL

G35 ~ mixed  
 G36 ~ mixed  
 G37 ~ mixed  
 G38 ~ mixed  
 G39 ~ mixed  
 G41 ✓ supported  
 G42 ✓ supported  
 G44 ~ mixed  
 G51 × failed

## FAITHFULNESS SCORECARD

|                        |   |                   |   |
|------------------------|---|-------------------|---|
| A mapping              | 1 | F identifiability | 1 |
| B assumptions          | 1 | G ground truth    | 1 |
| C adequacy             | 1 | H comparative     | 1 |
| D unfitted predictions | 1 | I transfer        | 0 |
| E interventional       | 1 |                   |   |

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

## CAVEATS

One hopper has  $\geq 5$  hop-days, and hopping read-outs total about 50 items. The test rules out  $\kappa \geq 0.1$  per e-fold of reads, not H48's small per-read pull. The #focus domain has two residents. #best had three stayers in #35–#37, which makes  $D$  and the interior occupancy noisy there. The day bootstrap for  $\kappa_R$  is liberal (size up to 0.18). Statement-level geometry depends on the embedding model; hopper positions agree to 0.05–0.25 between bge and gte. The “situational” reading is post hoc.

## ROUND 2

Measure on-entry and on-exit switching times per statement with ledger call times. Use NE42's merged week as the all-hopper limit.

- **Essence.** what sets the width of the boundary between two content domains in a swarm, and whether bridging agents widen it.
- **H102-R1** Situational vs carried state at the statement level.
- **H102-R2** The #35 room episodes (03-19/20).
- **H102-R3** Larger dose: NE42's merged week (#40).
- **H102-R4** Field as work: replace the kickoff-text direction with DQ4 repo labels per room (as H100-R3).